

Akash Rane , MS

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SUMMARY

Data Engineer with 3+ years of experience building batch and real-time data pipelines across identity, compliance, financial, and industrial IoT domains. Skilled in Python, SQL, Spark, Kafka, MQTT, and cloud data services, with experience developing ETL workflows, streaming ingestion systems, data-quality frameworks, and analytics-ready data models. Proven ability to integrate structured and unstructured data, support machine learning workflows, and deploy reliable data products for operational and business decision-making.

EDUCATION

Pace University, Seidenberg School of Computer Science and Information Systems

New York, USA

Master's in Computer Science | GPA: 3.89/4.00

Dec 2025

SPPU - Savitribai Phule Pune University

Pune, India

Bachelor of Engineering in Computer Engineering | **Concentration:** Data Science | **GPA: 8.18/10**

Jul 2022

TECHNICAL SKILLS

Languages & Query: Python (Pandas, Scikit-learn, NumPy), SQL, JavaScript,

Data & Pipelines: Apache Spark, ETL/ELT design, batch & streaming ingestion, Kafka, MQTT, OPC-UA, data validation & observability.

Cloud Platforms: AWS, Google Cloud Platform (GCP), cloud-based data services.

Visualization & Monitoring: Grafana, Streamlit, Plotly, Node-RED, Matplotlib.

DevOps & Infrastructure: Linux/UNIX, CI/CD workflows, Git, distributed systems fundamentals (partitioning, fault tolerance, scalability).

AI & ML Applied: Scikit-learn, XGBoost, Prophet, SHAP, LangChain, RAG / LLM-based retrieval, GitHub Copilot.

PROFESSIONAL EXPERIENCE

Shoptaki

New York

Data Engineer

Feb 2025 - Present

- Designed and built an end-to-end data pipeline using Python and Pandas to ingest, transform, validate, and standardize structured and unstructured identity data, producing analytics-ready datasets that powered an automated ML-based KYC verification workflow that routes each case to auto-reject, human review, or auto-verify reducing manual review to only ambiguous cases.
- Developed scalable document-processing pipelines to parse, chunk, normalize, and index structured and unstructured identity records, enabling efficient retrieval, traceability, and downstream analytics over compliance datasets.
- Implemented reusable data-quality and feature-engineering workflows for validation, cleaning, deduplication, and transformation of heterogeneous identity data, improving dataset reliability and supporting downstream identity classification and fraud-detection models.
- Designed feature extraction and transformation workflows supporting downstream analytics and machine learning models while ensuring data integrity across heterogeneous datasets.

Optify Industrial Solutions Pvt. Ltd.

Pune, India

Data Analyst & Business Intern

Jul 2022 - Dec 2023

- Implemented Python data engineering workflows to process high-frequency industrial sensor data, improving the availability and reliability of operational datasets used for equipment monitoring and maintenance analytics.
- Engineered real-time ingestion pipelines using OPC-UA and MQTT to stream sensor telemetry from industrial control systems into storage and Grafana/Node-RED dashboards, ensuring reliable low-latency delivery for continuous equipment monitoring.
- Built a batch processing layer that aggregated and transformed high-frequency telemetry into structured datasets, generating daily equipment-performance reports that accelerated production and maintenance decision-making.
- Led a one-month on-site production deployment as the primary technical lead, configuring industrial data integrations, troubleshooting real-time data pipelines, and collaborating with client stakeholders to achieve successful end-to-end system integration and production sign-off.

PROJECTS

Global Supply Chain Control Tower

Jan 2026

- Built a reusable data-processing pipeline that normalized country and logistics attributes, validated relational joins, engineered delivery and margin-risk metrics, and modeled supply-chain data into Fact Orders, Dim Geography, Dim Customers, Dim Products, and Dim Date tables.
- Created an end-to-end control tower with executive KPIs, interactive geographic and network visualizations, delay-risk exploration, forecasting, operational recommendations, what-if simulations, and automated executive reporting.
- Trained and integrated a Random Forest model using shipping mode, market, product department, scheduled delivery time, and logistics-performance indicators to estimate late-delivery risk at the individual-order level.

FinStream - Real - Time Financial Analytics Platform

Dec 2025

- Developed REST and WebSocket data services that integrated external financial-data sources and delivered live quotes, historical price series, market indices, and news to a responsive analytics application.
- Designed a batch quote-processing layer that separated cached and expired symbols, executed asynchronous upstream requests in parallel, and returned available cached records during provider failures to improve data availability.
- Optimized live-data delivery by introducing time-based caching and reducing polling frequency, limiting duplicate API traffic while supporting continuous portfolio, watchlist, and market-dashboard updates.

F1 Position and Pit-Stop Strategy Analytics

Mar 2025

- Built a multi-source ingestion workflow integrating race results, pit-stop records, weather feeds, racing schedules, and lap-level telemetry from Ergast, Open-Meteo, FastF1, and Selenium-based extraction.
- Cleaned, joined, and transformed race, driver, tire, weather, and pit-stop datasets covering the 1950–2024 Formula 1 seasons, engineering features such as lap-time variation, pit duration, tire usage, and driver behavior.